

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer:

The large difference between training MSE (0.85) and testing MSE (5.20) indicates that the neural network is overfitting the training data. The model performs very well on training data but fails to generalize to unseen testing data.

Possible Reasons:

1. **Overfitting:** The model has learned noise and specific patterns from the training data.
2. **Insufficient Training Data:** Small datasets may not represent all possible student score patterns.
3. **Complex Neural Network:** Too many hidden layers or neurons can increase model complexity.
4. **Lack of Regularization:** Absence of dropout or L1/L2 regularization may lead to overfitting.
5. **Data Distribution Difference:** Training and testing data may have different characteristics.
6. **Noisy Data:** Errors or outliers in the dataset can affect model performance.

Corrective Measures:

1. Apply Dropout to reduce overfitting.
2. Use L1/L2 Regularization to control model complexity.
3. Increase the amount of training data.
4. Use Early Stopping to prevent excessive training.
5. Reduce the number of hidden layers or neurons.
6. Perform proper data preprocessing and normalization.
7. Use Cross-Validation to evaluate model generalization.

Conclusion:

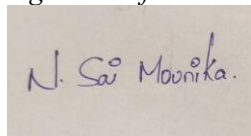
Since the testing MSE (5.20) is much higher than the training MSE (0.85), the model is likely overfitting. Regularization techniques, more training data, and simpler network architecture can improve generalization and reduce testing error.

Code of Conduct:

I Narra Sai Mounika Reg no: 192425367 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in purple ink that reads "N. Sai Mounika."

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided